Curriculum Vitae, M. Haque

Masudul Haque
Max-Planck Institute for Physics of
Complex Systems (MPI-PKS)
Nöthnitzer Strasse 38
01187 Dresden, Germany.

haque@pks.mpg.de
Tel. office +49-351-871-1109
Tel. mobile +31-6-1116-0834
http://www.pks.mpg.de/~haque/

Employment & Education

- **Current institute:** (since October 2006)

  Max-Planck Institute for the Physics of Complex Systems (MPI-PKS), Dresden, Germany.

  Currently Staff Scientist at MPI-PKS.

  Former position title: *Distinguished PKS Postdoctoral Fellow*.
  Held one of three such fellowships at MPI-PKS, until September 2009.
- Nov. 2003 to Sept. 2006: Postdoctoral researcher, Utrecht University, the Netherlands. Employed in cold-atom theory group led by H. T. C. Stoof.
- Ph.D., October 2003. Rutgers, State University of New Jersey.
  Advisor: Andrei E. Ruckenstein.
  Thesis title: *Interaction Effects in the Weakly Repulsive Bose Gas*.
- Graduate student and teaching assistant, 1997 to 2003.
  Rutgers, State University of New Jersey; Department of Physics & Astronomy.
- Lecturer, 1996-97. Physics Department, Shahjalal University, Bangladesh.
- M.Sc., 1996; B.Sc., 1995. Shahjalal University, Bangladesh.

Research Interests

- **Area:** Condensed Matter Theory.
  Collective phenomena in electronic, nanoscale and cold-atom systems.
- **Specific topics of published research:** Cross-discipline application of concepts from *Quantum Information Theory* (*e.g.*, entanglement measures) to condensed matter issues.

  Non-equilibrium phenomena; collective quantum dynamics.

  Topological order and unconventional states in many-particle systems.
  Fractional quantum Hall states. Frustrated magnets.

  Trapped atomic fermions and bosons — polarized fermion pairing; Feshbach resonances; BCS-BEC crossover; vortex dynamics in Bose condensates.

  Wigner crystallization. Pomeranchuk instabilities.
  Laser-excited semiconductors; excitons. Nonlinear dynamics.

Organization activities

Workshop in summer 2010

Title: Quantum information concepts for condensed matter problems. (QICCMP)
Co-organizers: Ian Affleck (UBC, Canada); Ulrich Schollwöck (LMU, Munich).
Details: 100^+ participants. Two-week programme (June 14–25, 2010).
Location: Max Planck Institute (MPI-PKS), Dresden, Germany.
Website: http://www.pks.mpg.de/∼qiccmp10/

Other initiatives & responsibilities

- Currently organizing a semi-weekly meeting on "Cold atoms & BEC" in Dresden.
- Organized the condensed matter theory seminar sequence at Utrecht, 2004–2006.
- Organized a lecture series on "1D & Exact Integrability" in Utrecht, spring 2006.
- As PhD student, administered large courses with several hundred students.

...... *etc.*

Teaching and Supervision

Citations refer to publication list.

- **Supervision:** Supervised junior researchers, for papers [5, 8, 7, 9, 11, 12, 15, 16, 25].
  Supervising summer research projects for visiting undergraduates,
  e.g., two in summer 2009, one leading to paper [5]. One in summer 2010.
  Guiding junior physicists in several ongoing projects.
- **1997-2001:** Teaching Assistant, Physics & Astronomy Dept, Rutgers.
  Teaching physics "recitations" and laboratories; course administration.
- **1996-1997:** Lecturer, Dept of Physics, Shahjalal University, Bangladesh.
  Full-time teaching job, similar to teaching in a U.S. 4-year college.

Grants & Fellowships

- *Grants* for collaboration visits, from Research Networking Programmes of the European Science Foundation (ESF) (programme acronym in brackets):

  ⋆ from Dresden to Paris; Dec. 2009 [INSTANS] & Feb. 2010 [INTELBIOMAT]
  ⋆ Dresden to Amsterdam; two visits, April 2009 and Nov. 2006. [INSTANS]
  ⋆ Dresden to Bristol; October/November 2008. [HFM]
  ⋆ Utrecht to U.K. (Cambridge & Birmingham); May 2006. [INSTANS]

  Papers [6, 11, 13, 14, 15] resulted partly from these visits.
- *Excellence Fellowship*, from Rutgers Graduate School Dean, 2002.

Early Independence in Research

- **Early independent publications:** One paper as Ph.D. student with co-students [24] ; one sole-author paper in 2005 [20]; both on topics unrelated to supervisors' interests.
- **Independent collaborations during 1st postdoc:**
  Initiated external collaborations (1) with K. Schoutens at Amsterdam,
  (2) with J. Quintanilla at Rutherford Lab and A. J. Schofield at Birmingham.
  Both collaborations led to later publications [7, 11, 13, 14, 15, 16].
- **Current work in Dresden:** Pursuing self-chosen directions of research.
  → Involves several collaborations, both on-site and long-distance.
  → Conceived & led research leading to papers [7, 9, 12, 25], and ongoing followups.

Talks and Seminars

Invited presentations in Conferences & Workshops

- ESF Research Conference (June 2010, Obergugl, Austria): *Quantum Engineering of States and Devices: Theory and Experiments.*
  Talk title: *Edge-locking and Quantum Control in 1D lattice models.*
- ICTP workshop (May 2009, Trieste, Italy): *Research Frontiers in Ultracold Atoms.*
  Talk title: *The Frontier of the Few: few-particle, few-vortex, and few-site surprises.*
- Dresden Workshop (March 2008): *Chaos and Collectivity in Many-Body Systems.*
  Talk title: *A vortex dipole in a trapped 2D condensate.*
- Amsterdam summer workshop (July '07): *Low-D Quantum Condensed Matter 2007.*
  Talk title: *Quantum information concepts for many-particle physics.*

...... *etc.*

Seminars at various institutes and departments (month/year in parenthesis):

Oxford (03/10); Munich LMU (02/10); Barcelona [Casteldefels] ICFO (01/10);
Paris Jussieu [LPTMC] (12/09); Grenoble ILL (11/09);
Dresden MPI-PKS (09/09, 11/06); Bristol (11/08, 09/07); ENS Paris (02/08);
M.I.T. (09/07); Florida Tallahassee (09/07); Rutgers (09/07, 12/05);
Düsseldorf (03/07); Groningen (01/07); Utrecht (11/06); Amsterdam (11/06);
Cambridge [Cavendish] (01/06); Rutherford Lab [ISIS] (11/05); Karlsruhe (07/05);
Frankfurt (07/05); ETH Zurich (12/04); *etc.*

Biographical Data

- **Birth:** 01/01/1974. Shariatpur, Bangladesh.
- **Citizenship:** Bangladesh.

Publication List

- **Overview:**
  - Papers marked with ▲ were coauthored solely with junior scientists or peers, supervised or led by myself.
  - Paper 6 is a review article.
  - Citations: 250+, in January 2010.
    For selected papers, individual citation counts (January 2010) are noted below.

- **Published:**

1. *Self-similar spectral structures and edge-locking hierarchy in open-boundary spin chains*. — M. Haque, Phys. Rev. A, **82**, 012108 (2010). arXiv:0906.0996.
2. *Disentangling Entanglement Spectra of Fractional Quantum Hall States on Torus Geometries*. — A. M. Laeuchli, E. J. Bergholtz, J. Suorsa, and M. Haque; Phys. Rev. Lett. **104**, 156404 (2010).
3. *Entanglement Scaling of Fractional Quantum Hall states through Geometric Deformations*. — A. M. Laeuchli, E. J. Bergholtz, and M. Haque; New Journal of Physics **12**, 075004 (2010).
4. *Interaction induced fractional Bloch and tunneling oscillations* . R. Khomeriki, D. O. Krimer, M. Haque, and S. Flach; Phys. Rev. A **81**, 065601 (2010).
5. *Finite-rate quenches of site bias in the Bose-Hubbard dimer*. T. Venumadhav, M. Haque, and R. Moessner; Phys. Rev. B **81**, 054305 (2010).
6. *Entanglement between particle partitions in itinerant many-particle states*. M. Haque, O. S. Zozulya, and K. Schoutens; arXiv:0905.4024. J. Phys. A: Math. Theor. **42**, 504012 (2009).
7. *Entanglement signatures of Quantum Hall phase transitions*. ▲ O. Zozulya, M. Haque, and N. Regnault; Phys. Rev. B **79**, 045409 (2009).
8. *Edge-localized states in quantum one-dimensional lattices*. R. A. Pinto, M. Haque, and S. Flach; Phys. Rev. A **79**, 052118 (2009).
9. *Entanglement and level crossings in finite frustrated ferromagnetic chains*. ▲ M. Haque, J. N. Bandyopadhyay, and V. Ravi Chandra; Phys. Rev. A **79**, 042317 (2009).
10. *Probing topological order in quantum Hall states using entanglement calculations*. M. Haque; AMS Contemp. Math. **482**, 213 (2009).
11. *Particle partitioning entanglement in itinerant many-particle systems*. O. Zozulya, M. Haque, and K. Schoutens; Phys. Rev. A **78**, 042326 (2008).
12. *A vortex dipole in a trapped two-dimensional Bose condensate*. ▲ W. Li, M. Haque and S. Komineas; Phys. Rev. A **77**, 053610 (2008).

13. *Symmetry-breaking Fermi surface deformations from central interactions in two dimensions.*
J. Quintanilla, M. Haque, and A. J. Schofield; Phys. Rev. B, **78**, 035131 (2008).

14. *Pomeranchuk instability: symmetry breaking and experimental signatures.*
J. Quintanilla, C. Hooley, B. J. Powell, A. J. Schofield, and M. Haque;
Physica B: Condensed Matter, **403**, 1279 (2008).

15. *Bipartite entanglement entropy in fractional quantum Hall states.*
O. Zozulya, M. Haque, K. Schoutens, and E. H. Rezayi;
Phys. Rev. B **76**, 125310 (2007). ★ 15^+ citations.

16. *Entanglement entropy of fermionic Laughlin states.*
M. Haque, O. Zozulya, and K. Schoutens; Phys. Rev. Lett. **98**, 060401 (2007).
★ 30^+ citations.

17. *Trapped fermionic clouds distorted from the trap shape due to many-body effects.*
M. Haque & H. T. C. Stoof; Phys. Rev. Lett., **98**, 260406 (2007).
★ 10^+ citations.

18. *Deformation of a Trapped Fermi Gas with Unequal Spin Populations.*
G. B. Partridge, Wenhui Li, Y. A. Liao, R. G. Hulet, M. Haque, and H. T. C. Stoof;
Phys. Rev. Lett. **97**, 190407 (2006).
★ 100^+ citations.

19. *Pairing of a trapped resonantly interacting fermion mixture with unequal spin populations.* — M. Haque & H. T. C. Stoof; Phys. Rev. A **74**, 011602 (2006).
★ 60^+ citations.

20. *Ring-shaped luminescence pattern in biased quantum wells studied as a steady state reaction front.* — M. Haque; Phys. Rev. E **73**, 066207 (2006).

21. *Squeezing in the weakly interacting uniform Bose-Einstein condensate.*
M. Haque & A. E. Ruckenstein; Phys. Rev. A, **74**, 043622 (2006).

22. *Ultracold superstring in atomic boson-fermion mixtures.*
M. Snoek, M. Haque, S. Vandoren, and H. T. C. Stoof;
Phys. Rev. Lett. **95**, 250401 (2005).
★ 10^+ citations.

23. *Dynamics of a molecular Bose-Einstein condensate near a Feshbach resonance.*
M. Haque & H. T. C. Stoof; Phys. Rev. A **71**, 063603 (2005).

24. *Structural Transition of Wigner Crystal on Liquid Substrate.* ▲
M. Haque, I. Paul and S. Pankov; Phys. Rev. B **68**, 045427 (2003).

25. *Survival benefits in mimicry: a quantitative framework.* ▲
A. Mikaberidze & M. Haque; J. Theor. Biol. **259**, 462 (2009).

26. *Anomaly in the normalization constant of the (^{3}He,p) reaction.*
M. Haque *et. al.*, Nuovo Cimento A, **111A**, 1131 (1998).

27. *^{64}Cu levels from the ^{62}Ni(^{3}He,p) reaction at 18 MeV.*
A. K. Basak *et. al.*, Phys. Rev. C **56**, 1983 (1997).

- **Publicly available preprints (submitted or unpublished)**

28. *Three-vortex configurations in trapped Bose-Einstein condensates*.
J. A. Seman, E. A. L. Henn, M. Haque, R. F. Shiozaki, E. R. F. Ramos, M. Caracanhas, C. Castelo Branco, G. Roati, K. Magalhaes, V. S. Bagnato, arXiv:0907.1584.
29. *Transition Temperature of Dilute Weakly Repulsive Bose Gas*.
M. Haque & A. E. Ruckenstein, cond-mat/0212590.
30. *Weakly Non-ideal Bose Gas: Comments on Critical Temperature Calculations*.
M. Haque, cond-mat/0302076.